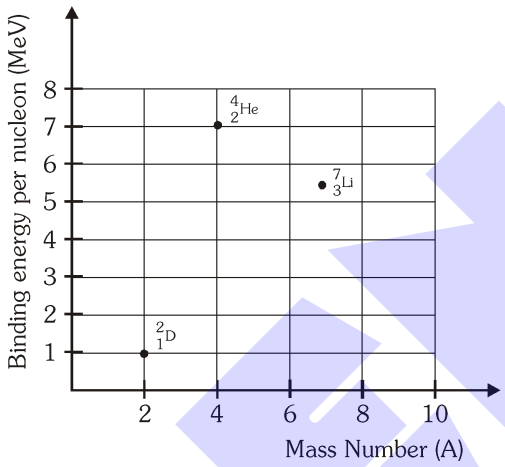
SECTION-I

Single Correct Answer Type **11 Q. [3 M (–1)]**

1. The positions of $^{2}_{1}D, ^{4}_{2}He$ and $^{7}_{3}Li$ are shown on the binding energy curve as shown in figure.

The energy released in the fusion reaction. $^{2}_{1}D + ^{7}_{3}Li \rightarrow 2\ ^{4}_{2}He + ^{1}_{0}n$



(A) 20 MeV (B) 16 MeV (C) 8 MeV (D) 1.6 MeV

2. The radionuclide $^{11}_{6}C$ decays by β^+ emission.

Given that $m(^{11}_{6}C) = 11.011434$ u

$m(^{11}_{5}B) = 11.009305$ u

$m_e = 0.000548$ u, $1u = 931.5$ MeV/c^2

The Q-value of this decay process is :-

(A) 0.962 MeV (B) 0.962×10^3 MeV (C) 0.962 eV (D) 0

3. A source contains two phosphorous radionuclides $^{32}_{15}P$ $(T_{1/2} = T_1)$ and $^{33}_{15}P$ $(T_{1/2} = T_2)$. Initially 10% of the decays come from $^{33}_{15}P$. How long one must wait until 90% do so ?

(A) $t = \dfrac{4\ell n3}{\ell n2\left(\frac{1}{T_1} - \frac{1}{T_2}\right)}$ (B) $t = \dfrac{4\ell n3}{\ell n2\left(\frac{1}{T_1} + \frac{1}{T_2}\right)}$ (C) $t = \dfrac{2\ell n3}{\ell n2\left(\frac{1}{T_1} - \frac{1}{T_2}\right)}$ (D) None of these

4. For a substance the average life for α-emission is 1620 years and for β emission is 405 years. After how much time the 1/4 of the material remains by simultaneous emission :-

(A) 648 years (B) 324 years (C) 449 years (D) 810 years

5. Tritium ($_1^3H$), an isotope of hydrogen with a half life of 12.3 year, is produced in the upper atmosphere by cosmic rays and is intimately mixed with the hydrogen in air. In order to determine the age of a bottle of wine found in an ancient cave, the tritium in the wine is measured and found to be 6.9% of that found in new wine. How old is the wine in bottle?

(A) 59.20 years (B) 47.5 years (C) 37.5 years (D) 53.2 years

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6. An unstable nucleus decays in three different modes, each mode having a different half life T_1, T_2, T_3($T_1 >> T_2 >> T_3$). The overall half–life of a sample will be given by–

(A) $T \approx T_1$ (B) $T \approx T_3$ (C) $T \approx \frac{T_1 + T_2 + T_3}{3}$ (D) $T \approx T_2$

7. The half-life of substance X is 45 years, and it decomposes to substance Y. A sample from a meteorite was taken which contained 2% of X and 14% of Y by quantity of substance. If substance Y is not normally found on a meteorite, what is the approximate age of the meteorite?

(A) 270 years (B) 135 years (C) 90 years (D) 45 years

8. A radioactive sample decays by β-emission. In first two seconds 'n' β-particles are emitted and in next 2 seconds, '0.25n' β-particles are emitted. The half life of radioactive nuclei is

(A) 2 sec (B) 4 sec (C) 1 sec (D) None of these

9. The radionuclide $^{11}_{6}C$ dacays by β^+ emission.

Given that $m(^{11}_{6}C) = 11.011434$ u

$m(^{11}_{5}B) = 11.009305$ u

$m_e = 0.000548$ u, $1u = 931.5$ MeV/c^2

The Q-value of this decay process is :-

(A) 0.962 MeV (B) 0.962×10^3 MeV (C) 0.962 eV (D) Zero

10. The nuclei of a certain radioactive element are produced at a constant rate of α per second. This element has a decay constant λ. Initially N_0 nuclei of the element are present. The number N of nuclei of this element at time t is

(A) $\frac{1}{\lambda}[\alpha - (\alpha - N_0\lambda)e^{-\lambda t}]$ (B) $\frac{1}{\lambda}[\alpha - N_0\lambda)e^{-\lambda t}$

(C) $(\alpha - N_0)\, e^{-\lambda t}$ (D) $\alpha\, N_0\, (1 - e^{-\lambda t})$

11. The electrostatic energy of Z protons uniformly distributed throughout a spherical nucleus of radius R is given by $E = \frac{3}{5}\frac{Z(Z-1)e^2}{4\pi\varepsilon_0 R}$ **[JEE Advance-2016]**

The measured masses of the neutron, $^{1}_{1}H$, $^{15}_{7}N$ and $^{15}_{8}O$ are 1.008665 u, 1.007825 u, 15.000109 u and 15.003065 u respectively. Given that the radii of both the $^{15}_{7}N$ and $^{15}_{8}O$ nuclei are same, $1u = 931.5$ MeV/c^2 (c is the speed of light) and $\frac{e^2}{(4\pi\varepsilon_0)} = 1.44$ MeV fm. Assuming that the difference between the binding energies of $^{15}_{7}N$ and $^{15}_{8}O$ is purely due to the electrostatic energy, the radius of either of the nuclei is ($1fm = 10^{-15}$m)

(A) 2.85 fm (B) 3.03 fm (C) 3.42 fm (D) 3.80 fm

Multiple Correct Answer Type **6 Q. [4 M (–1)]**

12. A nucleus has a radius of 7.2×10^{-15} m. When an α-decay takes place from this nucleus, ratio of number of neutrons and number of protons in the daughter nucleus becomes $\frac{65}{41}$:-

(A) Parent nucleus is $_{84}$Po. (B) Daughter nucleus is $_{82}$Pb.

(C) Mass number of daughter nucleus is 216. (D) Mass number of parent nucleus is 216.

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13. A radioactive sample has a half life of 40 seconds. When its activity is measured 80 seconds after the begining, it is found to be 6.932×10^{18} dps. During this time total energy released is 6×10^{8} joule ($\ell n2 = 0.6932$) :-

(A) The initial number of atoms in the sample is 1.6×10^{20}

(B) The initial number of atoms in the sample is 1.6×10^{21}

(C) Energy released per fission is 5×10^{-13} J

(D) Energy released per fission is $\frac{5}{3} \times 10^{-13}$ J

14. Choose the **CORRECT** option(s) :

(A) Strong nuclear force is a spin dependent force.

(B) Strong nuclear force between an electron and proton is same as between a proton and neutron.

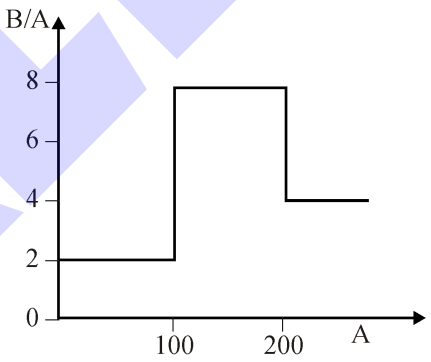
(C) Strong nuclear force is a short range force.

(D) Strong nuclear force is always attractive in nature.

15. A radioactive sample decays by three modes simultaneously. Half lives corresponding to these modes are in G.P. and half life of sample is 10 years. When the sample decays by the mode having largest half life it takes 70 years for the sample to become half. If sample decays exclusively by other modes possible values of half life is :

(A) 28 years (B) 35 years (C) 14 years (D) 17.5 years

16. Assume that the nuclear binding energy per nucleon (B/A) versus mass number (A) is as shown in the figure. Use this plot to choose the correct choice(s) given below : **[JEE 2008]**



(A) Fusion of two nuclei with mass numbers lying in the range of $1 < A < 50$ will release energy

(B) Fusion of two nuclei with mass numbers lying in the range of $51 < A < 100$ will release energy

(C) Fission of a nucleus lying in the mass range of $100 < A < 200$ will release energy when broken into two equal fragments

(D) Fission of a nucleus lying in the mass range of $200 < A < 260$ will release energy when broken into two equal fragments

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17. Mark out the **CORRECT** statement(s).

(A) Higher binding energy per nucleon means the nucleus is more stable.

(B) If the binding energy of nucleus were zero, then it would spontaneously break apart.

(C) Binding energy of a nucleus can be –ve.

(D) Binding energy of a nucleus is always +ve.

Linked Comprehension Type (1 Para × 3Q.) (1 Para × 2Q.) [3 M (-1)]

(Single Correct Answer Type)

Paragraph for Question No. 18 to 20

Polonium (P_0^{210}) emits ${}_2\alpha^4$ particles and is converted into lead (${}_{82}Pb^{206}$). This reaction is used for producing electric power in a space mission: P_0^{210} has half life of 138.6 days. (Given masses of the nuclei) P_0^{210} = 209.98264 amu, Pb^{206}= 205.97440, ${}_2\alpha^4$ = 4.00260 amu)

18. Assuming an efficiency of 10% of the thermoelectric machine, how much P_0^{210} is required to produce 1.2×10^7 J of electric energy per day at the end of 693 days?

(A) 1.0 gm (B) 2.0 gm (C) 3.0 gm (D) 0.5 gm

19. What is the initial amount of P_0^{210} ?

(A) 16 gm (B) 8 gm (C) 32 gm (D) 64 gm

20. Find the initial activity of the material.

(A) 4.57×10^{21} per day (B) 3.57×10^{21} per day

(C) 2.28×10^{21} per day (D) 1.785×10^{21} per day

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Paragraph for Questions 21 and 22

The β-decay process, discovered around 1900, is basically the decay of a neutron (n). In the laboratory, a proton (p) and an electron (e^-) are observed as the decay products of the neutron. Therefore, considering the decay of a neutron as a two-body decay process, it was predicted theoretically that the kinetic energy of the electron should be a constant. But experimentally, it was observed that the electron kinetic energy has a continuous spectrum. Considering a three-body decay process, i.e. $n \rightarrow p + e^- + \bar{\nu}_e$, around 1930, Pauli explained the observed electron energy spectrum. Assuming the anti-neutrino $(\bar{\nu}_e)$ to be massless and possessing negligible energy, and the neutron to be at rest, momentum and energy conservation principles are applied. From this calculation, the maximum kinetic energy of the electron is 0.8×10^6 eV. The kinetic energy carried by the proton is only the recoil energy.

21. If the anti-neutrino had a mass of 3 eV/c^2 (where c is the speed of light) instead of zero mass, what should be the range of the kinetic energy, K, of the electron? **[JEE 2012]**

(A) $0 \le K \le 0.8 \times 10^6 eV$ (B) $3.0eV \le K \le 0.8 \times 10^6 eV$

(C) $3.0eV \le K < 0.8 \times 10^6 eV$ (D) $0 \le K < 0.8 \times 10^6 eV$

22. What is the maximum energy of the anti-neutrino? **[JEE 2012]**

(A) zero (B) much less than 0.8×10^6 eV

(C) Nearly 0.8×10^6 eV (D) Much larger than 0.8×10^6 eV

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Matching List Type (4 × 4) 1Q.[3 M (–1)]

23. Match List I of the nuclear processes with List II containing parent nucleus and one of the end products of each process and then select the correct answer using the codes given below the lists:

[JEE Advance-2013]

	List I		List II
P.	Alpha decay	1.	${}^{15}_{8}O \rightarrow {}^{15}_{7}N + ...$
Q.	β^+ decay	2.	${}^{238}_{92}U \rightarrow {}^{234}_{90}Th + ...$
R.	Fission	3.	${}^{185}_{83}Bi \rightarrow {}^{184}_{82}Pb + ...$
S.	Proton emission	4.	${}^{239}_{94}Pu \rightarrow {}^{140}_{57}La + ...$

Codes:

	P	Q	R	S
(A)	4	2	1	3
(B)	1	3	2	4
(C)	2	1	4	3
(D)	4	3	2	1

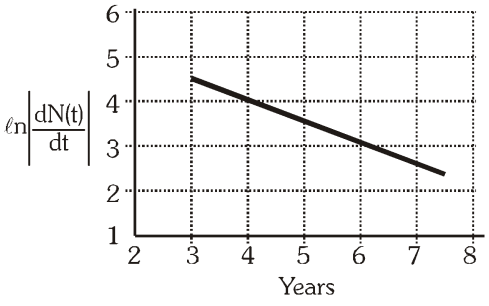
SECTION-III

Numerical Grid Type (Ranging from 0 to 9) 4 Q. [4 M (0)]

24. A radioactive sample contains two radio nucleoids A and B having decay constant λ hr^{-1} and 2λ hr^{-1}. Initially 25% of total decay comes from A. How long (in hr) will it take before 75% of total decay comes from A. [Take $\lambda = \ell n\, 3$]

25. A small quantity of solution containing ^{24}Na radionuclide (half life 15 hours) of activity 1.0 microcurie is injected into the blood of a person. A sample of the blood of volume 1 cm^3 taken after 5 hours shows an activity of 296 distintegration per minute. Determine the total volume of blood in the body of the person. Assume that the radioactive solution mixes uniformly in the blood of the person.
(1 Curie = 3.7×10^{10} disintegrations per second)

26. To determine the half life of a radioactive element, a student plots a graph of $\ell n\left|\frac{dN(t)}{dt}\right|$ versus t. Here $\frac{dN(t)}{dt}$ is the rate of radioactive decay at time t. If the number of radioactive nuclei of this element decreases by a factor of p after 4.16 years, the value of p is **[JEE-2010]**



27. Geiger counter reading of a radioactive sample is initially 6800 counts per min. The same sample gives a reading of 425 counts per min 10 hrs later. The sample's half life of α hr. Fill 2α in OMR sheet.

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SECTION-IV

Matrix Match Type (4 × 5) **1 Q. [8 M (for each entry +2(0)]**

28. Column I shows possible inferences on events given in column II.

	Column I		**Column II**
(A)	Not possible	(P)	Conversion of a free proton to a neutron.
(B)	Possible	(Q)	A neutron with kinetic energy 15 eV is incident on a stationary hydrogen atom in ground state such that excitation of H atom occurs.
(C)	Release of energy	(R)	Conversion of a neutron to proton outside nucleus.
(D)	A chargeless particle carrying momentum is emitted.	(S)	A nuclei emits electromagnetic radiation without emitting α or β particles.
		(T)	A thermal particle causing fission of nucleus.

Subjective Type **1 Q. [4 M (0)]**

29. A neutron of kinetic energy 65 eV collides inelastically with a singly ionized helium atom at rest . It is scattered at an angle of 90° with respect of its original direction.

(Given : Mass of he atom = 4×(mass of neutron), ionization energy of H atom =13.6 eV)

(i) Find the allowed values of the energy of the neutron & that of the atom after collision.

(ii) If the atom gets de-excited subsequently by emitting radiation, find the frequencies of the emitted radiation.

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GR # MODERN PHYSICS-II — ANSWER KEY

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SECTION-I

Single Correct Answer Type — 11 Q. [3 M (–1)]

1. Ans. (B) 2. Ans. (A) 3. Ans. (A) 4. Ans. (C) 5. Ans. (B)
6. Ans. (B) 7. Ans. (B) 8. Ans. (C) 9. Ans. (A) 10. Ans. (A)
11. Ans. (C)

Multiple Correct Answer Type — 6 Q. [4 M (–1)]

12. Ans. (A, B, D) 13. Ans. (B, C) 14. Ans. (A,C) 15. Ans. (B,D) 16. Ans. (B, D)
17. Ans. (A,B,D)

Linked Comprehension Type (1 Para × 3Q.) (1 Para × 2Q.) [3 M (-1)]
(Single Correct Answer Type)

18. Ans. (A) 19. Ans. (C) 20. Ans. (A) 21. Ans. (D) 22. Ans. (C)

Matching List Type (4 × 4) — 1Q.[3 M (–1)]

23. Ans. (C)

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) — 4 Q. [4 M (0)]

24. Ans. 2 25. Ans. 6 26. Ans. 8 27. Ans. 5

SECTION-IV

Matrix Match Type (4 × 5) — 1 Q. [8 M (for each entry +2(0)]

28. Ans. (A) → (P,Q) ; (B) → (R,S,T) ; (C) → RST (D) RST

Subjective Type — 1 Q. [4 M (0)]

29. Ans. (i) Allowed values of energy of neutron = 6.36 eV and 0.312 eV ;
Allowed values of energy of He atom = 17.84 eV and 16.328 eV ,
(ii) 18.23×10^{14} Hz , 9.846×10^{15} Hz , 11.6×10^{15} Hz

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